Karlsruhe Institute of Technology

Test, Validation, Analysis

Title of the Thesis

Bachelor's Thesis
of

Name Surname

at the Department of Informatics
Institute of Information Security and Dependability
Test, Validation and Analysis (TVA)

Reviewer:	Prof. Dr.
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Ich versichere wahrheitsgemäß, die Arbeit selbstständig verfasst, alle benutzten Hilfsmittel vollständig und genau angegeben und alles kenntlich gemacht zu haben, was aus Arbeiten anderer unverändert oder mit Abänderungen entnommen wurde sowie die Satzung des KIT zur Sicherung guter wissenschaftlicher Praxis in der jeweils gültigen Fassung beachtet zu haben.

Karlsruhe, den ??. ?????? 2023

Zusammenfassung

Neben dem Titel ist die Zusammenfassung der wichtigste Teil Ihrer Arbeit, da die meisten Leser nur den Titel und die Zusammenfassung lesen werden. Ihr Ziel ist es, den Rest der Arbeit für potenzielle Leser zu bewerben. Dazu erklären Sie kurz, worauf Sie sich in der Arbeit konzentrieren. Mit einer irreführenden Zusammenfassung verpassen Sie interessierte Leser und locken vielleicht sogar Leser mit falschen Annahmen über Ihre Arbeit an, die dann bald aufhören zu lesen. Die Zusammenfassung sollte sowohl das allgemeine Gebiet als auch die interessantesten Erkenntnisse der Arbeit beschreiben. Es ist entscheidend, das richtige Maß an Abstraktion und Länge zu finden. Eine Zusammenfassung besteht in der Regel aus einem Absatz, der deutlich kürzer ist als die Einleitung.

Kurzfassungen folgen in der Regel dem gleichen Aufbau. Sie beginnen mit der Beschreibung des Forschungsgebiets sowie des allgemeinen und des spezifischen Problems, auf das Sie sich konzentrieren. Dann skizzieren Sie, wie Sie das Problem in Form von Konzepten und Bewertungen angehen. Abschließend stellen Sie die interessantesten Erkenntnisse dar, die Sie gewonnen haben, und begründen, warum diese für das Forschungsgebiet relevant sind.

. . .

Abstract

Besides the title, the abstract is the most important part of your thesis, as most readers will only read title and abstract. Your goal is to advertise the rest of the thesis for potential readers. For that, you briefly explain what you are focusing on in the thesis. With a misleading abstract, you will miss interested readers and maybe even attract readers with wrong assumptions about your work which will stop reading soon. The abstract should describe the general area as well as the most interesting insights of the work. It is crucial to find the right level of abstraction and length. An abstract typically consists of one paragraph that is significantly shorter than the introduction.

Abstracts typically follow the same structure. You start by describing the research area as well as the general and the specific problem you are focusing on. Then, you outline how you approach the problem in terms of concepts and evaluations. Finally, you close with the most interesting insights that you gained and why they are relevant for the research area.

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Acronyms

TVA	Institute for Test, Validation and Analysis of Software Intensive Systems
KIT	Karlsruhe Institute of Technology
KASTEL	Institute of Information Security and Dependability

1. Introduction

. . . to be continued . . .

. . .

Goal of this Thesis

Structure of the Thesis

2. Basics

The basics must be described to the extent that a reader understands the problem and the solution to the problem. In order not to describe too much to describe, this can be written towards the end of the work. Everything must be included that is necessary for the understanding of the further work and is not an own contribution.

...

3. Design

In this chapter, the detailed description of the own solution approach is given. Alternative solutions should be discussed and design decisions should be presented.

...

3.1 Summary

At the end, if necessary, the most important results should be summarized again in ***a*** *short paragraph.*

...

4. Example Chapter

This chapter gives you some examples how to include graphics, create tables, or include code listings. But first, we start with a short description how you can efficiently cite in LaTeX. The following footnote contains an url which is defined in */utility/urls.tex* of this documents folder structure.[1] The footnote leads to the website of Institute for Test, Validation and Analysis of Software Intensive Systems (TVA). TVA is a research group at Karlsruhe Institute of Technology (KIT) in the Institute of Information Security and Dependability (KASTEL). And now you have already seen that this template makes use of acronyms. With Acronyms you can define abbreviations for terms you use frequently in your thesis. In this template the acronyms are defined in */essentials/acroonyms.tex*. The first time you reference an acronym using
`ac{acronym}` LaTeXwill use the long form and the abbreviation provided in the definition of the acronym as seen before. Every time thereafter LaTeXuses the abbreviation provided in the acronym definition (e.g, TVA, KIT, and KASTEL). See TVA, KIT, and KASTEL are shown only in their abbreviated forms. You can force the long by using the
`acl{acronym}` command (e.g, Institute for Test, Validation and Analysis of Software Intensive Systems, Karlsruhe Institute of Technology, and Institute of Information Security and Dependability).

4.1 Citation

There are several types of literature. The most citations are workshop and conference papers. Please use the inproceedings-tag for those citations (e.g., [KAK09]). You should have short-hands for workshop and conference names to be sure the naming is consistent and uniform (see our BibTeX files how to do that).

Slightly different are articles published in journals (e.g., [KG06]). Make sure you that the volume and number-tags are present and that no inproceeding is tagged as article or vice versa.

[1] https://tva.kastel.kit.edu/

You might want to take a look at the example BibTeX file to find out how to cite books [CE00], technical reports [Kan+90], websites [Coq09], PhD theses, or master theses [Beu03; Ros09].

4.2 Formulas

There are different types of mathematical environments to set formulas. The equation $E = m \cdot c^2$ is an inline formula. But you can also have formulas at a separate line (see 4.1).

$$P = \big(\mathcal{A} \Rightarrow (\mathcal{B} \Leftrightarrow \mathcal{C}) \wedge (\mathcal{B} \Leftrightarrow \mathcal{D})\big) \wedge (\mathcal{B} \Rightarrow \mathcal{A}) \wedge (\mathcal{C} \Rightarrow \mathcal{A}) \wedge (\mathcal{D} \Rightarrow \mathcal{A}) \tag{4.1}$$

If you need multiple lines that are aligned to each other, you might want to use the following code.

$$\begin{aligned} & \text{GraphLibrary} \\ \wedge\ & (\text{GraphLibrary} \Rightarrow \text{Edges}) \wedge (\text{Edges} \vee \text{Algorithms} \Rightarrow \text{GraphLibrary}) \\ \wedge\ & (\text{Edges} \Leftrightarrow \text{Directed} \vee \text{Undirected}) \wedge (\neg\text{Directed} \vee \neg\text{Undirected}) \\ \wedge\ & (\text{Algorithms} \Leftrightarrow \text{Number} \vee \text{Cycle}) \\ \wedge\ & (\text{Cycle} \Rightarrow \text{Directed}). \end{aligned}$$

4.3 Graphics

In 4.1, we give a small example how to insert and reference a figure.

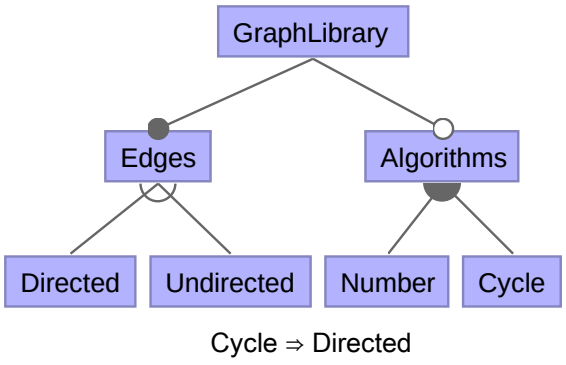


Figure 4.1: A feature model representing a graph product line

4.4 Tables

4.1 on the facing page shows the result of a simple tabular environment.

4.5 Code Listings

In 4.1 on the next page, we give an example of a source code listing.

Group Type	Propositional Formula
And	$(P \Rightarrow C_{k_1} \wedge \ldots \wedge C_{k_m}) \wedge (C_1 \vee \ldots \vee C_n \Rightarrow P)$
Or	$P \Leftrightarrow C_1 \vee \ldots \vee C_n$
Alternative	$(P \Leftrightarrow C_1 \vee \ldots \vee C_n) \wedge \text{atmost1}(C_1, \ldots, C_n)$

Table 4.1: Mapping a feature model to a propositional formula

```
1  class A extends Object {
2      A() { super(); }
3  }
4  class B extends Object {
5      B() { super(); }
6  }
7  class Pair extends Object {
8      Object fst;
9      Object snd;
10     Pair(Object fst, Object snd) {
11         super(); this.fst=fst; this.snd=snd;
12     }
13     Pair setfst(Object newfst) {
14         return new Pair(newfst, this.snd);
15     }
16 }
```

Listing 4.1: Java source code

5. Implementation

Most theses in computer science are accompanied by tool support written by the author of the thesis. Such tools enable an empirical evaluation or simply serve as a proof-of-concept. In particular, tools are typically not the ultimate goal in research, but often necessary to evaluate whether proposed concepts solve real problems. Hence, it is common to write about the tool in a dedicated chapter.

The tool chapter has several goals. For supervisors, it typically helps to estimate the implementation effort of a thesis and problems faced during development. For other students, the chapter serves as the documentation of the tool support. That is, students that extend the tool support will use this chapter to get an overview on the architecture and learn from failed attempts. As researchers are typically rather interested in concepts or evaluations, this dedicated chapter on tool support helps to remove clutter from other chapters. Nevertheless, researchers may be interested to read why tool support has been build the way it is and why it is build on certain existing tools or libraries. Write the chapter such that it useful for researchers, students, and supervisors.

. . .

6. Evaluation

...

7. Related Work

...

8. Conclusion and Outlook

Summarize the contribution of the entire work and highlight the importance of the results

Outlook on further development of the results

(No more subsections!)

...

Bibliography

[Beu03] Danilo Beuche. "Composition and Construction of Embedded Software Families". PhD thesis. Germany: University of Magdeburg, 2003.

[CE00] Krzysztof Czarnecki and Ulrich W. Eisenecker. *Generative Programming: Methods, Tools, and Applications*. ACM Press/Addison-Wesley, 2000.

[Coq09] The Coq Development Team. *The Coq Proof Assistant*. Website. Available online at http://coq.inria.fr/; visited on November 9th, 2009. Sept. 2009.

[KAK09] Christian Kästner, Sven Apel, and Martin Kuhlemann. "A Model of Refactoring Physically and Virtually Separated Features". In: Denver, CO, USA: ACM, Oct. 2009, pp. 157–166.

[Kan+90] Kyo C. Kang et al. *Feature-Oriented Domain Analysis (FODA) Feasibility Study*. Tech. rep. CMU/SEI-90-TR-21. Software Engineering Institute, 1990.

[KG06] Cory J. Kapser and Michael W. Godfrey. "Supporting the Analysis of Clones in Software Systems: A Case Study". In: 18.2 (2006), pp. 61–82.

[Ros09] Malte Rosenthal. "Alternative Features in Colored Featherweight Java". Diplomarbeit. Germany: University of Passau, July 2009.

A. Appendix

Appendix